ARUNKUMAR S

AI/ML Engineer — Computer Vision & Generative AI

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SUMMARY

AI/ML Engineer with 2+ years of experience delivering production Generative AI and Computer Vision systems for defense, automotive, and commercial vehicle clients including HAL, VTI (Volvo Trucks India), VECV, TMPV (JLR), and TVS. Build air-gapped GenAI solutions (RAG, TTS, LoRA/PEFT fine-tuning of multilingual LLMs, LangChain/LangGraph agents) alongside real-time vision pipelines (YOLOv8, pose estimation, VLM reasoning, TensorRT on local RTX GPUs). Experienced in end-to-end system development, from model training and optimization to FastAPI/Flask service design and on-premises deployment, with a focus on reliable, production-ready delivery.

EXPERIENCE

Jr. Software Engineer

06/2024 – Present

GGS Information Services Pvt. Ltd. — *Bangalore, India*

- **HAL - Offline RAG Platform:** Designed and built a fully offline Retrieval-Augmented Generation platform (Ollama, Qdrant, Podman, WSL2) for a client with strict data-residency requirements, covering PDF ingestion, semantic chunking, embedding generation, and context-aware retrieval.
- **HAL - Offline TTS Engine:** Built a fully offline text-to-speech engine with word-level synchronized highlighting and an intelligent text-normalization layer for engineering abbreviations, units, and symbols; a custom caching layer cut repeated-inference latency and kept playback responsive.
- **VTI (Volvo Trucks India) - Crash Impact Detection:** Trained and deployed a YOLOv8/OpenCV model for real-time vehicle damage assessment, reaching ~87% mAP across three severity classes and cutting manual inspection effort by ~50%.
- **VECV - 3D-to-2D Documentation Automation:** Built a Blender + Python pipeline that programmatically explodes CAD assemblies and generates callouts, converting 3D models into structured SVG line-art and reducing manual technical-illustration time by over 70%.
- **TMPV (JLR) - LLM Fine-Tuning:** Applied LoRA/PEFT fine-tuning on Qwen2.5-Instruct (multilingual LLM) for domain-specific document understanding, lifting response accuracy ~12% over the base model on automotive service-query benchmarks.
- **TVS - Assembly-Line Video Analytics (POC):** Built a real-time system to check that operators perform assembly steps in the correct order. YOLOv8-Pose (TensorRT-optimized) tracks operator pose to detect each action, Qwen3-VL confirms the step from key frames, and missed or out-of-order steps are logged and searchable by station and shift. Runs on an on-premises RTX GPU workstation.
- **Agentic Workflows:** Orchestrated LangChain and LangGraph agents across 4+ tools — ingestion, retrieval, reasoning, generation — into autonomous document Q&A pipelines for enterprise knowledge bases. Used LangSmith to trace, evaluate, and debug multi-step agent runs.
- **API & Platform:** Built and maintain three production Flask/FastAPI services, with centralized error handling and structured JSON I/O for reliable inference at scale.

PROJECTS

RAG-Powered GenAI Chatbot with Voice Interface — *Personal Project*

- Built a full-stack RAG chatbot in personal time using Flask, FAISS, LangChain and AWS Bedrock: PDF upload, text extraction and vector retrieval (~83% retrieval accuracy), with Bedrock Guardrails filtering unsafe inputs and responses
- Extended the chatbot with a full voice interface using AWS Polly (TTS) and Transcribe (STT), adding speech-based interaction on top of the retrieval pipeline.

Technologies: Python • AWS Bedrock • Bedrock Guardrails • LangChain • FAISS • Flask • AWS Polly • AWS Transcribe

EDUCATION

Bachelor of Technology, Information Technology

2020 – 2024

SRM Easwari Engineering College — *Chennai, India | CGPA: 8.54/10*

TECHNICAL SKILLS

Backend & APIs: Python, Flask, FastAPI, REST APIs, SQL, NumPy, Pandas, scikit-learn

Generative AI: RAG, Agentic AI, LangChain, LangGraph, LangSmith, FAISS, Qdrant, Ollama, AWS Bedrock, LoRA/PEFT, VLMs (Qwen3-VL), NLP, Prompt Engineering, Vector Databases, Hugging Face

AI-Native / Agentic Development: GitHub Copilot, Cursor, Claude Code

Computer Vision: PyTorch, TensorFlow, YOLO (Darknet, Ultralytics), YOLOv8-Pose, OpenCV, CNN, LSTM, Transformers, MediaPipe, OpenPose, OCR (PaddleOCR, Tesseract)

Automation: Blender Python API, SVG Generation, CAD Automation

Cloud (AWS): Bedrock, Bedrock Guardrails, Polly, Transcribe, API Gateway, Textract

DevOps & Deployment: Docker, Podman, WSL2, Git, GitHub

Model Optimization: ONNX, ONNX Runtime, TensorRT, Model Quantization (INT8), GGUF, llama.cpp, On-Premise LLM Inference